

## Coding & Solution

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|----------------------|----------------------------------------------------------|
| <b>Date</b>          | 02 October 2026                                          |
| <b>Team ID</b>       | 6ab7f6f8c23107aaaa72871e                                 |
| <b>Project Name</b>  | FitBuddy - AI Fitness Plan Generator using Gemini Models |
| <b>Maximum Marks</b> | 5 Marks                                                  |

### Coding & Solution

The implemented workflow connects the FastAPI frontend/backend, Gemini AI modules and SQLite persistence into a single browser-based user experience.

#### Core Route Logic

/generate-workout receives user fields, builds a validated UserInput object, calls Gemini Pro for the workout plan and Gemini Flash for the tip, stores the results and renders result.html. /submit-feedback retrieves the original plan, calls the update function and saves the revised result.

#### AI Integration

generate\_workout\_gemini() uses Gemini 1.5 Pro for structured 7-day plans;  
 generate\_nutrition\_tip\_with\_flash() uses Gemini Flash for a concise goal-aligned tip;  
 update\_workout\_plan() sends the original plan plus feedback back to Gemini Pro.

#### Persistence

save\_user(), save\_plan(), update\_plan(), get\_original\_plan() and get\_user() support SQLite persistence through SQLAlchemy ORM.

#### Frontend Integration

index.html collects profile details; result.html shows user data, workout plan, nutrition tip and feedback form; all\_users.html displays stored records for admin/coach review.